

Freight Rate Prediction Challenge

1. Approach to Validating and Splitting Train/Test Data

The training data spans January October 2025, and exploratory analysis showed that the average posted rate drifts over the course of the year (rising from January into June, dipping in July August, then recovering into September and October). Because the real prediction task is forecasting rates for future, unseen dates, a conventional random train/test split was rejected: shuffling rows across months would let the model see information from the same time period it is being evaluated on, and would overstate how well it generalizes to genuinely new dates. The split strategy instead follows chronological, month based validation throughout.

1.1 Rolling chronological tuning folds

For hyperparameter and model family selection, the training data is grouped into calendar months. Starting several months before the final holdout month, each candidate model is fit on every row dated strictly before a given month and evaluated only on that month, never on data the model could not have seen yet at prediction time. This produced three tuning folds in the current run:

- Fold 1: train on Jan Jun 2025, validate on Jul 2025
- Fold 2: train on Jan Jul 2025, validate on Aug 2025
- Fold 3: train on Jan Aug 2025, validate on Sep 2025

Twenty five candidate configurations were compared this way (Ridge, Extra Trees, and Histogram Gradient Boosting, each fit on both the raw target and a log1p transformed target, across several hyperparameter settings, plus one variant trained only on non outlier rows). Each candidate's score is the mean RMSE across the three folds, which rewards configurations that generalize consistently across different months rather than performing well on only one.

1.2 Untouched final month as an honest holdout

October 2025 is withheld from tuning entirely; it is never used to select a model family, hyperparameters, or the outlier handling policy. Once the best performing

candidate is chosen from the rolling folds above, it is refit on all data before October and scored a single time against October, which was never part of the selection process. This is the closest approximation to genuine out of sample performance available within the training data, and it is reported separately from the tuning score so that the two are never conflated.

In the current run, the winning configuration (Histogram Gradient Boosting on a $\log_{10}$ target) scored a mean MSE of 388,491.88 (RMSE 623.29) across the three tuning folds recorded in `model_comparison.csv` as `mean_mse` / `mean_rmse` for every candidate and MSE 420,090.03 (RMSE 648.14) / MAE 115.39 / R^2 0.820 on the untouched October holdout. MSE is reported alongside RMSE because it is the quantity the model is actually fit to minimize, while RMSE restates it in the same dollar units as `posted_rate` for easier interpretation (RMSE is simply the square root of MSE). The higher holdout error compared to the tuning score is expected and is treated as the more trustworthy estimate, since the tuning score is somewhat optimistic from having been used to pick the winner.

1.3 Leakage safe feature preparation

All feature engineering (city coordinate correction, distance sanity checks, weight/market/quote imputation, calendar features) is fit exclusively on the training split and then applied unchanged to the validation and December sets. Medians used for imputing missing weight, market index, and quote signal values are computed only from training rows; the same fitted values are reused rather than recomputed on validation or December data, which would otherwise leak information back from the evaluation set into the features.

Target derived columns (rate per mile, log rate, outlier flags) are excluded from the model's feature matrix entirely, since they are only computable when the target is already known and would not be available for genuinely new loads.

1.4 Final production model

After the untouched month check confirms the selected configuration, the same configuration is refit one final time on the complete training set (all 48,000 rows, January October 2025) so that the deployed model benefits from every available observation, including October, before generating predictions for the validation set and the December chart inputs.

2. December Prediction Chart

The chart below was produced by the competition's provided scorer (score.py), which first validates the submission format 12,000 validation predictions matching the required load IDs, and 31 December predictions covering every day of December 2025 with pickup, delivery, distance, equipment, and weight held fixed so that only the date varies and then renders the chart from the validated December predictions.

Fixed inputs for every row: Lexington to Fort Wayne, 360 miles, Dry Van, 32,000 lb. Only the date changes across the 31 points, isolating the model's learned seasonal/day of month pricing pattern for this route.

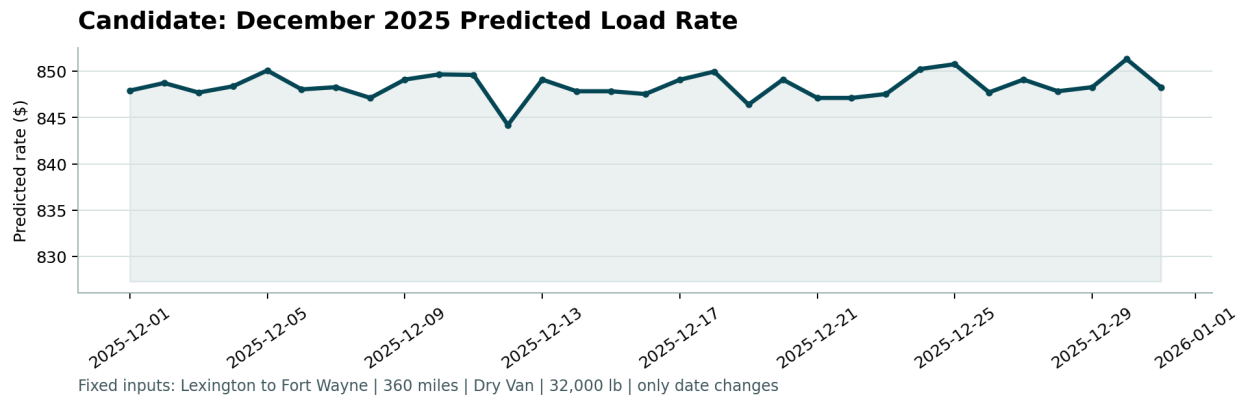


Figure 1. Candidate December 2025 predicted load rate (score.py output, scorer_results/candidate_december.png).